

MD TAZBIRUL HAQUE

Senior AI Engineer · Agentic Systems & LLM Applications · AI Platform Engineering · Developer Productivity AI

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PROFESSIONAL SUMMARY

Senior AI Engineer and enterprise software practitioner with 7+ years of production engineering experience, specialising in agentic systems, multi-agent orchestration, LLM applications, RAG pipelines, and AI platform engineering. Architected and delivered a GitHub Copilot and Claude-based AI-assisted development PoC at CTS EVENTIM that progressed through the full AI product lifecycle — PoC to MVP to production — with 3 AI products now live in customer-facing production environments and 5+ engineering teams actively implementing features. Currently engineering production AI agents, multi-agent workflows, tool-calling orchestrations, RAG pipelines with semantic search and knowledge retrieval, context management, agent memory, workflow state management, and human-in-the-loop systems. Established AI evaluation, model evaluation, AI observability, and LLM monitoring frameworks to ensure production reliability, hallucination mitigation, and output quality governance after deployment. Led organisation-wide AI adoption: mentored 5+ engineering teams on GitHub Copilot and Claude AI workflows, promoted AI-first development practices, and established human-reviewed AI-assisted development processes with responsible AI usage and AI governance guidelines. Deep expertise in enterprise security (OAuth2/OIDC, RBAC, SSO via Keycloak), AI observability (OpenTelemetry), and cloud-native production deployment (Docker, Kubernetes, Helm, CI/CD). Brings end-to-end ownership — from architecture through customer deployment — combined with the engineering discipline to make AI systems secure, observable, and reliably production-grade.

CORE COMPETENCIES

AI Engineering & Agentic Systems · AI Agents · Multi-Agent Systems · Agentic Workflows · Agent Orchestration · Tool Calling · MCP · RAG Pipelines · LLM Applications
Developer Productivity AI · AI-Assisted Development · AI-Assisted Code Review · Workflow Automation · AI Product Development · AI Transformation Leadership · PoC → MVP → Production → Organisation-wide Adoption
Human-in-the-Loop Workflows · AI Governance · AI Quality Assurance · AI Observability · Model Evaluation · LLM Monitoring · Hallucination Mitigation · Guardrails · Responsible AI
Enterprise Security & Identity Management · OAuth2/OIDC · RBAC · SSO · JWT · Security-first AI Systems · GDPR-aware Architecture
Cloud-Native & Distributed Systems · Microservices · Event-Driven Architecture · Kubernetes · Docker · CI/CD · OpenTelemetry · Distributed Tracing · Production Observability
Enterprise Backend Engineering · C#/.NET · Python · REST APIs · GraphQL · RabbitMQ · Azure Service Bus · MS SQL Server · PostgreSQL

TECHNICAL SKILLS

AI Engineering: AI Agents, Multi-Agent Systems, Agentic Workflows, Agent Orchestration, Tool Calling, MCP, RAG Pipelines, Embeddings, Semantic Search, Knowledge Retrieval, Context Management, Agent Memory, Workflow State Management, Human-in-the-Loop Workflows, Prompt Engineering, Prompt Optimisation, Chunking Strategies, AI Evaluation, Model Evaluation, AI Observability, LLM Monitoring, Hallucination Mitigation, Guardrails, AI Output Validation, AI Governance, AI Quality Assurance, Production AI Systems, Workflow Automation, AI Product Development, AI Automation, Prompt Versioning, Prompt Testing, Agent Evaluation, Evaluation Pipelines, Offline Evaluation, Online Evaluation, A/B Testing for AI Systems, Feedback Loops, Human Feedback Systems

AI Reliability & Evaluation: Agent Evaluation, Evaluation Pipelines, Feedback Loops, Grounded Generation, Output Verification, Retrieval Quality Evaluation, Inference Pipelines, Model Routing, Hybrid Search, Knowledge Indexing

LLM & AI Tools: Claude (Anthropic), LLM APIs, GitHub Copilot, Claude Code, AI-Assisted Development, AI-Assisted Code Review, Developer Productivity AI, Developer Enablement, Human-Reviewed AI Workflows, Responsible AI Usage, AI Adoption Leadership

Security & IAM: Keycloak, OAuth2/OIDC, RBAC, SSO, JWT, Secure API Design, Secrets Management, GDPR-aware Architecture, Security-first AI Systems, Data Classification

Observability & Monitoring: OpenTelemetry, AI Observability, LLM Monitoring, Distributed Tracing, Metrics, Diagnostics, Production Visibility, Error Handling, Performance Monitoring

Backend Engineering: C#, .NET, ASP.NET Core, Web API, GraphQL, Entity Framework, REST APIs, RabbitMQ, Azure Service Bus, Microservices, Event-driven Architecture

Cloud & DevOps: Azure DevOps, Azure Services, Azure Service Bus, Docker, Kubernetes, Helm, GitLab CI/CD, YAML Pipelines, Containerisation, Serverless (conceptual), CI/CD

Databases: MS SQL Server (Stored Procedures, Indexes, Query Optimisation, Role-based Permissions), PostgreSQL, MongoDB, Entity Framework Core, Relational Data Modelling

Languages: C# (primary), .NET, SQL, Python (ML research + AI engineering), JavaScript, TypeScript (familiar)

Frontend: React, Angular, Blazor, MVC, REST API Integration

Testing & Quality: NUnit, Integration Testing, AI Evaluation Frameworks, AI Output Validation, Production Quality Standards, Code Reviews, Clean Code, Documentation

Methodologies: Agile Scrum, Technical Documentation, Legacy Modernisation, Cross-functional Collaboration, Swagger / OpenAPI, Jira, Confluence

PROFESSIONAL EXPERIENCE

Software Development Expert — AI & Backend Engineering · CTS EVENTIM AG & Co. KGaA

· Bremen, Germany · 2023 – Present

- **AI Product Lifecycle — PoC to Production:** Architected and delivered a GitHub Copilot and Claude-based AI-assisted development PoC that progressed through the full AI product lifecycle — PoC → MVP → production deployment → organisation-wide adoption. 3 AI products are now live in customer-facing production environments; 5+ engineering teams are actively implementing features on the platform.
- **Agentic Systems & Multi-Agent Orchestration:** Engineering production AI agents, multi-agent systems, and agentic workflows using planner-executor patterns, tool calling, agent orchestration, sub-agent coordination, agent memory, workflow state management, and human-in-the-loop workflows. Designing agent coordination and task decomposition strategies to improve reliability, observability, and maintainability at enterprise scale.
- **RAG Pipelines & Knowledge Retrieval:** Designing and implementing production RAG pipelines including embedding generation, chunking strategies, semantic search, hybrid search, knowledge indexing, retrieval architecture, retrieval quality evaluation, context engineering, and grounding LLM outputs in internal knowledge sources. Applying hallucination mitigation strategies and output validation to ensure accuracy, factual grounding, and user trust across deployed AI systems.
- **Prompt Engineering & Output Quality:** Applying prompt engineering, prompt optimisation, prompt testing, and prompt versioning to improve agent output quality; implementing guardrails, grounded generation techniques, AI output validation, output verification, and hallucination mitigation to ensure production-safe LLM behaviour. Treating AI-generated outputs as untrusted input to be hardened and validated before use.
- **AI Evaluation, Observability & Monitoring:** Establishing AI evaluation, agent evaluation, and model evaluation frameworks with evaluation pipelines, feedback loops, human feedback systems, offline evaluation, online evaluation, and production quality gates to monitor agent performance degradation and support continuous improvement of deployed AI systems. Implementing AI observability and LLM monitoring using OpenTelemetry-based instrumentation to ensure agentic systems remain reliable, governable, and auditable long after launch.
- **AI Reliability Engineering:** Designed inference pipelines, model routing strategies, and production monitoring approaches to improve scalability, reliability, and operational efficiency of AI-powered workflows while maintaining human oversight and governance standards.
- **AI Governance & Human Oversight:** Implementing AI governance standards: defining review processes for AI-generated code, establishing AI quality assurance checkpoints, and ensuring human supervision of AI systems remains central to all AI-assisted workflows. Enforcing responsible AI usage and data classification practices as production defaults.
- **Developer Productivity AI & Organisation-wide Enablement:** Led AI adoption across the engineering organisation after MVP delivery: mentored 5+ engineering teams on GitHub Copilot and Claude AI workflows, established developer enablement programmes, promoted AI-first development practices, and implemented human-reviewed AI-assisted development processes, AI governance standards, and responsible AI usage guidelines — driving measurable developer productivity improvements at enterprise scale.
- **Enterprise Backend & Legacy Modernisation:** Architected and delivered core backend systems as part of an enterprise modernisation program: migrated a legacy PowerBuilder module to C# / ASP.NET Core / .NET with REST APIs, GraphQL endpoints, and distributed service integrations — delivering production-grade systems while concurrently leading the AI engineering initiative.
- **Security-first Architecture:** Implemented enterprise IAM using Keycloak (OAuth2/OIDC, RBAC, SSO, JWT) as a security-first production standard; applied secure API design, secrets management, and GDPR-aware architecture across both backend and AI-facing systems.

- **Production Observability:** Established production observability and AI observability using OpenTelemetry: distributed tracing, LLM monitoring, metrics, diagnostic instrumentation, and error handling across distributed backend and AI services.
- **Infrastructure & Data:** Deployed and operated production services via Docker, Kubernetes, and Helm with CI/CD pipelines; implemented event-driven communication using RabbitMQ; designed and optimised MS SQL Server structures including stored procedures, role hierarchies, indexes, and query performance tuning.

Senior Full Stack Developer · Movingdots GmbH / Swiss Re / Powerfleet · Bremen, Germany · 2021 – 2023

- **Technical Ownership & Leadership:** Promoted to Senior Developer with end-to-end technical ownership of the COLORED/Coloride telematics product — leading architecture decisions, cross-functional delivery, production operations, and team coordination.
- **Applied ML & Model Evaluation (Master's Thesis):** Applied Python for supervised machine learning research: designed, trained, and evaluated a real-time crash detection algorithm using SVM (Support Vector Machines), ANN (Artificial Neural Networks), and Decision Trees with systematic hyperparameter optimisation on real telematics sensor data. Conducted model evaluation, performance benchmarking, and feature engineering — establishing the applied ML and model evaluation foundation now extended into production AI engineering.
- **Full Stack Development:** Developed full stack features with React and Angular frontends and C# / ASP.NET Core backend services integrated with PostgreSQL and MongoDB; implemented production observability with OpenTelemetry across distributed telematics services.
- **Cloud & DevOps:** Built and deployed services on Azure: Azure DevOps pipelines, Azure Service Bus for asynchronous event-driven messaging, Azure cloud infrastructure, and CI/CD deployment environment configuration.

Software Developer · Netzlab GmbH · Dortmund, Germany · 2020 – 2021

- Developed and maintained C# / .NET backend systems and web services; contributed to software architecture, backend development, bug fixing, maintenance, and technical documentation for production systems.

Software Developer · Computer Communication Ltd. · Bangladesh · 2017 – 2019

- **Enterprise Banking Software:** Built production-grade enterprise banking software for Citi Bank — cashing systems and reporting modules — using C# / .NET Core, MVC, Blazor, and MS SQL Server with strict data integrity, security, and reliability requirements.

SELECTED ACHIEVEMENTS

- **AI Product Lifecycle — End-to-End:** Led complete AI product lifecycle from concept to PoC, MVP, customer production, and organisation-wide adoption. 3 AI products live in customer-facing production; 5+ engineering teams enabled.
- **Agentic AI Platform Engineering:** Building production multi-agent systems, RAG pipelines, semantic search, tool-calling orchestrations, context management, and skill libraries — with real customer deployment outcomes at enterprise scale.
- **Developer Productivity AI & AI Transformation:** Mentored 5+ engineering teams on GitHub Copilot and Claude AI-assisted development; established human-reviewed AI development processes and AI governance guidelines across the engineering organisation.
- **AI Governance & Production Reliability:** Defined AI output validation, guardrails, hallucination mitigation, and human supervision standards to ensure production-safe, governable agentic systems with long-term reliability after deployment.
- **AI Reliability & Evaluation:** Established evaluation pipelines, retrieval quality controls, grounded generation techniques, feedback loops, and human approval workflows to improve trust, governance, and long-term reliability of production AI systems.
- **Enterprise Legacy Modernisation:** Delivered enterprise-scale migration from PowerBuilder to modern C# / .NET platform while concurrently leading AI engineering transformation — demonstrating senior-level multi-initiative ownership.

EDUCATION

M.Sc. Information Engineering and Computer Science · Hochschule Rhein-Waal, Germany

Thesis: AI-based Crash Detection in Telematics Systems — Real-time classification of vehicle crash events using supervised machine learning on sensor telemetry data.

Relevant Areas of Study:

Machine Learning (SVM, ANN, Decision Trees, Hyperparameter Optimisation) · Applied Python for Data Science & ML · Model Training, Evaluation & Benchmarking · Feature Engineering & Data Preprocessing · Signal Processing & Sensor Data Analysis · Software Engineering & Distributed Systems · Data Management & Database Systems · Algorithm Design & Optimisation · Research Methods & Scientific Writing

B.Sc. Computer Science and Engineering · Ahsanullah University of Science and Technology, Bangladesh

Relevant Areas of Study: Data Structures & Algorithms · Object-oriented Programming · Database Systems · Software Engineering · Computer Networks · Operating Systems · Web Technologies · Artificial Intelligence (Foundations)

LANGUAGES

English: Fluent **German:** Professional working proficiency (B1) **Bengali:** Native